12TH BIOLOGY · CHAPTER 18 · FULL CHAPTER NOTES

Inheritance

Topic-wise short and long questions with answers from the complete chapter

18.1 LAW OF SEGREGATION

Q.1 [Short] What is a true-breeding plant?

A plant which is true-breeding (pure-breeding) for a particular trait produces all the offspring with the same trait upon self-fertilization. Gregor Johann Mendel developed true-breeding varieties of pea plants.

Q.2 [Long] Explain Mendel's monohybrid cross and the law of segregation.

Mendel performed monohybrid crosses in pea plants. He crossed a true-breeding round-seeded plant with a wrinkled-seeded plant. All plants of the F_1 generation produced round seeds. He called the trait expressed in F_1 (round) **dominant** and the trait not expressed (wrinkled) **recessive**. When the F_1 plants were self-pollinated, 1/4 of the F_2 seeds were wrinkled, i.e., the ratio was 3 dominant : 1 recessive. This 3:1 ratio was really a disguised 1:2:1 ratio: 1/4 pure-breeding dominant, 2/4 not pure-breeding dominant and 1/4 pure-breeding recessive.

F_1 gametes (Rr × Rr)	R	r
R	RR (round)	Rr (round)
r	Rr (round)	rr (wrinkled)

Interpretation: Characters are controlled by "factors" (now called genes). Each individual is diploid and has two genes for a character. When the individual forms haploid gametes, each gamete receives just one gene copy (allele), which is selected randomly. This is known as the **law of segregation**.

Q.3 [Short] Define gene, allele and locus.

A **gene** is a sequence of nucleotides, as part of DNA, which codes for the formation of a polypeptide. The alternative forms of a gene, leading to alternative traits, are called **alleles**. The particular location of a gene on a chromosome is called the gene's **locus**.

Q.4 [Short] Differentiate between homozygous and heterozygous; genotype and phenotype.

When two gametes with the same alleles fuse, the offspring is **homozygous** (e.g., RR, rr). When gametes with different alleles fuse, the offspring is **heterozygous** (Rr). The combination of alleles in an individual is its **genotype** (the blueprint); the observable character is the **phenotype** (the visible outcome).

Q.5 [Short] Why did the recessive trait disappear in F_1 and reappear in F_2?

In the heterozygous F_1 (Rr), only the dominant allele is expressed, while the recessive allele is present but unexpressed. When F_1 plants form gametes, the alleles segregate. When two gametes carrying r fuse, the offspring is rr and the recessive trait reappears (in 1/4 of F_2).

Q.6 [Short] What is a testcross? What is its use?

A testcross is used to determine whether a dominant phenotype is homozygous or heterozygous. The dominant phenotype is crossed with a recessive phenotype. For example, a round-seeded plant may be RR or Rr. It is crossed with a wrinkled-seeded plant (rr).

- If it is RR, all offspring will be round (Rr).
- If it is Rr, the offspring will be round and wrinkled in a 1:1 ratio.

18.2 LAW OF INDEPENDENT ASSORTMENT

Q.7 [Long] Explain Mendel's dihybrid cross and the law of independent assortment.

Mendel studied two characters together: seed shape, round (R) dominant and wrinkled (r) recessive; and seed colour, yellow (Y) dominant and green (y) recessive.

1. He crossed pure-breeding round yellow plants (RRYY) with pure-breeding wrinkled green plants (rryy).
2. All F_1 seeds were round and yellow, with the genotype RrYy. These F_1 plants are called **dihybrids**.
3. He self-fertilized the F_1 plants. Each dihybrid produced four kinds of gametes: RY, Ry, rY and ry.
4. When 4 male × 4 female gametes combined, 16 combinations were formed in F_2.

F_1 gametes	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

F_2 phenotypic ratio: 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green.

The results showed that the genes for seed shape assort independently of the genes for seed colour. So Mendel proposed the **law of independent assortment**: "each pair of alleles assorts independently of other pairs of alleles during gamete formation".

Q.8 [Short] Does independent assortment change the 3:1 ratio of single characters?

No. Independent assortment of different genes does not affect the segregation of individual allele pairs. In the dihybrid F_2, round vs wrinkled seeds still show 12:4 = 3:1, and yellow vs green seeds also show 3:1.

Q.9 [Short] Give the modern re-statement of the law of independent assortment.

“Genes that are located on different chromosomes assort independently during meiosis.”

Q.10 [Long] How does independent assortment lead to variation? Relate it to the orientation of chromosomes in meiosis.

Independent assortment mixes alleles during the formation of gametes and so creates genetic diversity. It happens because homologous chromosome pairs line up **randomly** during meiosis. This random alignment causes different combinations of maternal and paternal alleles in gametes. In humans, it can produce over 8 million (2^{23}) possible allele combinations in one gamete, so every gamete is genetically unique. The variation increases further at fertilization, when a sperm and an egg combine their unique alleles to form a genetically unique zygote. This genetic diversity helps species adapt to changing environments.

Q.11 [Short] What are the limitations of the law of independent assortment?

1. **Linked genes:** Genes close together on the same chromosome are often inherited together, not independently (gene linkage).
2. **Chromosome crossover:** The exchange of genetic material between homologous chromosomes during meiosis changes the predicted ratio of gene combinations.
3. **Epistasis:** The effect of one gene is modified or suppressed by one or more unrelated genes, which changes the expected ratio.
4. **Pleiotropy:** One gene controls multiple characters, so these characters are linked and the expected phenotypic ratios are distorted.
5. **Environmental effect:** Many phenotypes can be affected by the environment.

Q.12 [Short] What is pleiotropy?

Pleiotropy is the phenomenon in which one gene controls multiple characters. These characters are therefore linked, and the expected phenotypic ratios of independent assortment are distorted.

Q.13 [Long] What is probability? Explain the rules of probability used in genetics.

Probability means how often a particular outcome occurs. The transfer of genes from one generation to the next is random, so the possible new allelic combinations are calculated by mathematical probability.

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Basic rule: Probability = number of times an event occurs ÷ total number of possible outcomes. Example: if wrinkled seeds appear 1,850 times out of 7,324 seeds, the probability is 1,850 ÷ 7,324 = 0.253 (about 1 in 4).

Product rule: The chance of two or more independent events happening together is found by multiplying their individual probabilities. Example: in Aa × Aa, each parent has a 1/2 chance of giving allele a, so the probability of aa = 1/2 × 1/2 = 1/4. In RrYy × RrYy, round yellow = 3/4 × 3/4 = 9/16.

Sum rule: For mutually exclusive events, the probability of either event is found by adding their individual probabilities. Example: in Aa × Aa, the dominant phenotype can occur in three ways (AA, Aa, aA), each with probability 1/4, so the total = 1/4 + 1/4 + 1/4 = 3/4.

Q.14 [Short] When are probability rules preferred over a Punnett square?

A Punnett square is suitable for one or two genes (e.g., RrYy × RrYy). For many genes (e.g., AaBbCcDdEe × AaBbCcDdEe) the Punnett square becomes too complex, so probability rules are used instead.

Q.15 [Short] Phenylketonuria and albinism are two recessive disorders which assort independently. If both parents are heterozygous for both traits, what is the chance of a child with (a) phenylketonuria, (b) albinism, (c) both traits?

Parents: PpAa × PpAa. (a) Pp × Pp gives pp = **1/4**. (b) Aa × Aa gives aa = **1/4**. (c) Both together (product rule) = 1/4 × 1/4 = **1/16**.

Q.16 [Short] What proportions of the offspring from a CC × Cc cross would be homozygous dominant, homozygous recessive and heterozygous?

CC gives only C gametes; Cc gives C and c. Offspring: 1/2 CC and 1/2 Cc. So homozygous dominant = **1/2**, heterozygous = **1/2**, homozygous recessive = **0**.

Q.17 [Short] A TtYy pea plant self-pollinates and one seed is picked at random. (a) What is the chance that it will produce a tall, green-seeded plant? (b) If it turns out to be tall and yellow-seeded, what is the chance that its genotype is TTYY?

(a) Tall (T_) = 3/4 and green (yy) = 1/4, so tall green = 3/4 × 1/4 = **3/16**.

(b) Tall yellow plants are 9 out of 16, and only 1 of these 9 is TTYY. So the chance = **1/9**.

18.3 DOMINANCE RELATIONS AND MULTIPLE ALLELES

Q.18 [Short] What is complete dominance?

In complete dominance, the dominant allele is fully expressed and the recessive allele is not expressed at all in the heterozygote, so no intermediate trait appears. Example: in pea plants, Yy seeds are yellow.

Q.19 [Long] Explain incomplete dominance with the example of the four-o'clock plant.

In incomplete dominance, neither allele is fully dominant over the other. The heterozygous individual shows a phenotype that is **intermediate** between the phenotypes of both parents.

Example: In Japanese four-o'clock plants, when a red-flowered plant (RR) is crossed with a white-flowered plant (WW), all the F_1 offspring are heterozygous and pink-flowered (RW). This shows that neither the allele for red (R) nor that for white (W) is completely dominant. When two pink-flowered plants (RW × RW) are crossed, the F_2 shows red (RR), pink (RW) and white (WW) flowers in a 1:2:1 ratio.

F_1 gametes (RW × RW)	R	W
R	RR (red)	RW (pink)
W	RW (pink)	WW (white)

Q.20 [Short] A pink-flowered four o'clock plant is crossed with a red-flowered plant. Find (a) the probability of red, (b) the probability of pink, (c) the ratio of pink to red.

RW × RR gives 1 RR : 1 RW. (a) Red = **1/2**. (b) Pink = **1/2**. (c) Pink : red = **1 : 1**.

Q.21 [Long] What is co-dominance? Explain with the MN blood group.

Co-dominance is the type of inheritance in which both alleles of a gene pair are fully expressed in a heterozygous individual. Both alleles work without blending and without one masking the other, so the products of both alleles appear together in the phenotype.

Example: the human MN blood group. Allele L^M produces M antigen and allele L^N produces N antigen. Both are codominant, so if both are present, both antigens appear on the red blood cells.

Genotype	Antigen	Blood group
L^ML^M	M	M
L^NL^N	N	N
L^ML^N	M and N	MN

Q.22 [Short] What does "L" stand for in the MN system? Why is it also called the MNS system?

"L" stands for Landsteiner and Levine, who discovered this system in 1927. It is also called the MNS system because the genes of the MN group and the S/s group are physically linked on the same chromosome.

Q.23 [Short] Differentiate between incomplete dominance and co-dominance.

Incomplete dominance	Co-dominance
In the heterozygous state, both genes blend their phenotypic effects.	In the heterozygous state, both genes fully express their effects.
The heterozygote shows an intermediate phenotype between the two parental phenotypes.	The heterozygote shows both parental phenotypes at a time.
Example: flower colour in Japanese 4 O'clock plant.	Example: human MN blood group and human AB blood group.

Q.24 [Short] What are multiple alleles? Give an example.

A multiple allelic system occurs when a gene has more than two alternative forms (alleles) in a population. This creates more variation in traits. Example: the human ABO blood group system is controlled by one gene (I) with three alleles: I^A, I^B and i. Each person has only two of these alleles.

Blood group	Genotypes	Antigen on RBCs	Antibodies in plasma
A	I^AI^A or I^Ai	Antigen-A	Anti-B
B	I^BI^B or I^Bi	Antigen-B	Anti-A
AB	I^AI^B	Antigen-A and antigen-B	None
O	ii	None	Anti-A and anti-B

18.4 BLOOD GROUP SYSTEMS

Q.25 [Short] What is a blood group system? Name the major systems.

A blood group system is a way of classifying blood based on the presence of specific antigens on red blood cells. The International Society of Blood Transfusion recognizes 43 blood group systems in humans. The most important is the ABO system and the second is the Rh system; matching of both is necessary for transfusion. Other major systems are:

Blood group system	Well-known antigens
MNS	M, N, S, s, U
Lutheran	Lu^a and Lu^b
Kell	K, k, Kp^a, Kp^b, Js^a, Js^b
Lewis	Le^a and Le^b
Duffy	Fy^a, Fy^b, Fy3

Q.26 [Long] Describe the ABO blood group system and its inheritance.

The ABO system was discovered by Landsteiner in 1901. He found two antigens on red blood cells: antigen-A and antigen-B. People with antigen-A have blood group A, those with antigen-B have blood group B, those with both have blood group AB, and those with neither have blood group O. Each person's plasma contains antibodies against the antigen he or she does not have (Landsteiner's rule).

Inheritance: The system is controlled by the gene I (iso-haem-agglutinogen), located on chromosome 9. It has multiple alleles: I^A, I^B and i. I^A produces antigen-A and I^B produces antigen-B; they are co-dominant over each other. Allele i produces no antigen and is recessive to both.

Blood group	Genotypes	Antigen on RBCs	Antibodies in plasma
A	I^AI^A or I^Ai	Antigen-A	Anti-B
B	I^BI^B or I^Bi	Antigen-B	Anti-A
AB	I^AI^B	Antigen-A and antigen-B	None
O	ii	None	Anti-A and anti-B

Q.27 [Short] State Landsteiner's rule of antigen-antibody relationship.

Each person's blood plasma contains antibodies against the antigen he or she does not have. Blood group A has anti-B antibodies and blood group B has anti-A antibodies. In the ABO system these antibodies do not require any exposure to antigens; they start forming from embryonic life.

Q.28 [Short] What is the secretor gene?

Some people secrete antigens A and B in body fluids, e.g., saliva, tears and urine. They possess a dominant secretor gene "Se" on chromosome 19.

Q.29 [Short] A man has blood group A and his wife has B. They have four children, each with a different blood group. How is it possible?

It is possible when both parents are heterozygous: $I^Ai \times I^Bi$.

Gametes	I^A	i
I^B	I^AI^B (AB)	I^Bi (B)
i	I^Ai (A)	ii (O)

Q.30 [Short] A woman with blood group B has a child with blood group O. What are the genotypes of the mother and child? Which genotypes could the father not have?

The child is **ii**, so the mother must carry i: she is $\mathbf{I^Bi}$. The father must also give i, so he can be I^Ai, I^Bi or ii. He could **not** be I^AI^A, I^BI^B or I^AI^B.

Q.31 [Short] Explain the principle of transfusion in the ABO system.

During transfusion between two incompatible blood groups, the recipient's antibodies attack the antigens on the donor's RBCs. This causes clumping (**agglutination**) of RBCs. It happens if a blood group A individual receives blood group B and vice versa, and if the donor is AB and the recipient is A, B or O.

- **O blood group** has no antigen-A or antigen-B, so it can be given to A, B, AB or O. O individuals are the universal donors of the ABO system.
- **AB blood group** has no anti-A or anti-B antibodies, so it can receive all blood types. AB individuals are the universal recipients of the ABO system.

Q.32 [Short] Why is the recessive allele 'i' more frequent in the population?

The allele i is recessive, so it can remain hidden in heterozygous people of blood groups A and B and pass to the next generation. Many different crosses produce it:

- O (ii) × O (ii): all offspring have blood group O (ii).
- $I^Ai \times I^Ai$: 25% offspring are O (ii).
- $I^Bi \times I^Bi$: 25% offspring are O (ii).
- $I^Ai \times I^Bi$: 25% offspring are O (ii).

That is why the allele i is more frequent in the population.

Q.33 [Long] Describe the Rh blood group system.

The Rh system is the second most important blood group system in humans. Its name comes from the Rhesus monkey, in which its antigen was first discovered by Landsteiner in the 1930s. It is based on the presence or absence of the **Rh-antigen** (Rh factor or antigen-D) on RBCs. Individuals having this antigen are **Rh-positive** and those lacking it are **Rh-negative**.

Inheritance: It is controlled by a gene D with two alleles. The dominant allele D forms the Rh-antigen; the recessive allele d does not. So genotypes DD and Dd are Rh-positive, and dd is Rh-negative.

Transfusion: Rh-negative people do not naturally have anti-Rh antibodies. But if an Rh-negative person receives Rh-positive blood, his body starts producing anti-Rh antibodies, which destroy the donated RBCs. These antibodies stay in the body for life.

Q.34 [Short] Why is O-negative the universal donor and AB-positive the universal recipient?

O-negative is the actual universal donor because its RBCs have neither any antigen of the ABO system (A or B) nor the antigen of the Rh system. So no antibody of the recipient can attack the donated RBCs.

AB-positive is the actual universal recipient because its plasma has neither anti-A and anti-B nor anti-Rh antibodies. So it can receive any donor's blood.

Q.35 [Long] Explain erythroblastosis foetalis and its prevention.

Erythroblastosis foetalis is a haemolytic disease of the foetus or new-born caused by Rh incompatibility between an **Rh-negative mother** and her **Rh-positive foetus**.

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Antigen–antibody reaction

1. An Rh-negative mother (dd) carries an Rh-positive baby (DD or Dd).
2. Foetal blood may enter the mother's blood. The foetal RBCs carry the Rh-antigen.
3. On exposure to these Rh-antigens, the mother produces anti-Rh antibodies.
4. These antibodies can cross the placenta and enter the foetal blood.
5. There, the mother's anti-Rh antibodies destroy the foetal RBCs. This leads to haemolytic anaemia and other serious complications.

Complications

In severe cases, the destruction of foetal RBCs can cause miscarriage or stillbirth. If the foetus survives, the liver and spleen become enlarged because they try to make more RBCs. The breakdown of RBCs produces bilirubin, which builds up and causes jaundice. High bilirubin can also damage brain cells.

Prevention and treatment

- When there are chances of Rh incompatibility, the mother is given an injection of **Rh-antiserum** (serum containing anti-Rh antibodies) during early pregnancy and immediately after birth. It destroys any of the baby's RBCs that have crossed into the mother's blood before they sensitize her immune system.
- The surviving baby's blood is replaced by Rh-negative blood that is free of anti-Rh antibodies.

Q.36 [Short] How can ABO incompatibility protect a baby against Rh incompatibility?

If an O-negative mother conceives an A-positive or B-positive baby, any foetal RBCs entering her blood are quickly destroyed by her already present anti-A or anti-B antibodies, before she can form anti-Rh antibodies.

Q.37 [Short] An Rh-negative woman is married to an Rh-positive man whose father was Rh-negative. Give the genotypes and the chance of erythroblastosis foetalis.

Woman: dd. Man's father: dd, so the man received d from him and is **Dd**. Cross dd × Dd gives 1/2 Dd (Rh-positive) and 1/2 dd (Rh-negative). So there is a **50% chance** that a child is Rh-positive and therefore at risk of erythroblastosis foetalis. (The risk is for an Rh-positive baby after the mother has been sensitized.)

18.5 POLYGENIC INHERITANCE AND EPISTASIS

Q.38 [Short] What is polygenic inheritance? Give examples of polygenic traits.

Polygenic characters are controlled by many gene pairs, not just one. In polygenic inheritance, all the alleles of many genes contribute to the same trait in an **additive** way, and their combined effect produces a single trait. Polygenic traits show a range of differences, i.e., **continuous variation**, in a population.

Examples: height, skin colour, weight and intelligence in humans, and grain colour in plants.

Q.39 [Short] Explain the inheritance of human skin colour.

Skin colour depends on the amount of melanin pigment. It is controlled by polygenes (about 60 genes); for understanding, three genes A, B and C are taken. Alleles A, B and C produce more melanin and alleles a, b and c produce less. If the parents are AABBCC (very dark) and aabbcc (very light), all their children will be AaBbCc (intermediate colour). If both parents are AaBbCc, their children will show a wide range of skin colours in the ratio 1 : 6 : 15 : 20 : 15 : 6 : 1 (from very dark to very light).

Q.40 [Long] Describe polygenic inheritance of grain colour in wheat.

Wheat grain colour shows continuous variation, from white to dark red. It is controlled by three polygenes: A, B and C. Each gene has a dominant allele (A, B, C) that produces red pigment and a recessive allele (a, b, c) that does not.

1. The genotype AABBCC produces the maximum red pigment, so the grains are dark red. The genotype aabbcc produces no pigment, so the grains are white.
2. When dark red wheat (AABBCC) is crossed with white wheat (aabbcc), the F_1 generation (AaBbCc) shows an intermediate red colour.
3. When F_1 plants are crossed, a wide range of colours appears in F_2. Out of 64, only one plant has the darkest red grains (AABBCC) and one has white grains (aabbcc). The other 62 plants show different shades of red.

Number of pigment-producing alleles	6	5	4	3	2	1	0
F_2 plants (out of 64)	1	6	15	20	15	6	1

Environmental factors like light, water and nutrients also influence the grain colour.

Q.41 [Short] Differentiate between allelic and non-allelic interactions.

In **allelic interactions**, alleles of the same gene interact with each other, e.g., incomplete dominance, codominance and multiple alleles. In **non-allelic interactions**, alleles of different genes (at different loci) interact with each other, e.g., epistasis.

Q.42 [Short] Define epistasis, epistatic allele and hypostatic allele.

Epistasis is the gene interaction in which an allele masks or modifies the effect of the alleles of other genes (present at other loci). It is a non-allelic interaction. The allele which masks or modifies the effect is called **epistatic**, and the allele whose effect is masked or modified is called **hypostatic**. For example, in humans, the allele of baldness is epistatic to the alleles of black and blonde hair.

Q.43 [Long] Explain epistasis in the coat colour of Labrador.

In Labrador dogs, coat colour (black, brown, yellow) depends on two genes.

1. **Gene B** controls pigment colour. Dominant B gives black, recessive b gives brown. So BB or Bb is black and bb is brown.
2. **Gene E** controls the deposition of pigment in fur. Dominant E allows pigment, but recessive e is epistatic to B or b: it blocks pigment deposition. So ee dogs have a yellow coat, whether B or b is present.

When two heterozygous black dogs (BbEe × BbEe) are crossed, the offspring are:

Genotypes	Fraction	Coat colour
B_E_	9/16	Black
bbE_	3/16	Brown
B_ee and bbee	3/16 + 1/16 = 4/16	Yellow

So the Mendelian 9:3:3:1 ratio is modified to **9 black : 3 brown : 4 yellow**.

Q.44 [Long] Explain epistasis in the flower colour of foxglove.

In foxglove plants, flower colour (dark red, light red, white) is controlled by two genes.

1. **Gene D:** dominant D produces dark red petals; recessive d produces light red petals.
2. **Gene W** (at another locus): dominant W limits the pigment to small spots, so the petals appear white from outside. Recessive w allows uniform pigment distribution.

When a heterozygous white-flowered plant (DdWw) is self-fertilized:

Genotypes	Fraction	Flower colour
D_W_ and ddW_	9/16 + 3/16 = 12/16	White
D_ww	3/16	Dark red
ddww	1/16	Light red

So the ratio is modified to **12 white : 3 dark red : 1 light red**.

Q.45 [Short] What is the Bombay phenotype?

The expression of ABO genotypes also depends on another gene, H. It makes the H substance, which is the precursor of the A and B antigens. If a person has I^A or I^B alleles but the genotype hh, no H substance is made, so no A or B antigen is produced. The person shows the O phenotype. This is called the Bombay phenotype because it was discovered in Bombay in 1952.

Q.46 [Short] Differentiate between polygenic inheritance and epistasis.

Both involve more than one gene. In polygenic inheritance, many genes act together, each with a small effect, and the result is continuous variation. In epistasis, one allele masks or modifies the effect of another gene, which usually gives discontinuous variation (distinct categories).

18.6 GENE LINKAGE AND CROSSING OVER

Q.47 [Short] State the chromosomal theory of inheritance.

In 1900, Karl Correns rediscovered Mendel's work and suggested the role of chromosomes in heredity. In 1902, Walter Sutton proposed the chromosomal theory of inheritance: (1) genes are found at specific locations on chromosomes; (2) the behaviour of chromosomes during meiosis explains Mendel's laws of inheritance.

Q.48 [Short] What is gene linkage?

In body cells, thousands of genes are present on a limited number of chromosomes. So many genes are arranged in a line on each chromosome. Genes present on the same chromosome are called **linked genes**, and this is known as **gene linkage**. Closely linked genes tend to stay together and do not assort independently during gamete formation. Genes on different chromosomes, or those far apart on the same chromosome, assort independently.

Q.49 [Short] How is gene linkage detected?

Gene linkage is studied by performing crosses and comparing parental combinations with recombinant (new) combinations in the offspring.

- If parental and new phenotypes appear in an equal 1:1:1:1 ratio, the genes are **not linked** and assort independently.
- If parental combinations are more frequent and new combinations are fewer, there is **incomplete linkage**: the genes are linked but can separate due to crossing over.
- If only parental combinations appear, there is **complete linkage**: the genes are very close and do not separate.

Q.50 [Short] Why can linkage be observed only when the number of progeny is large?

The larger the progeny size, the more likely it is to represent the actual frequency of parental and recombinant types. In a small progeny, the ratios may differ by chance, so linkage cannot be evaluated reliably.

Q.51 [Long] Explain Morgan's experiment on gene linkage in *Drosophila*.

Thomas Hunt Morgan studied two traits in *Drosophila*: wing length and abdomen width. The allele for long wings (L) is dominant over vestigial wings (l), and the allele for broad abdomen (B) is dominant over narrow abdomen (b).

1. He crossed a fly with long wings and broad abdomen with a fly with vestigial wings and narrow abdomen. All F_1 flies had long wings and broad abdomens.
2. He then crossed two F_1 flies.
3. **Expected result:** According to the law of independent assortment, F_2 should show a 9:3:3:1 ratio: 9 long broad and 1 vestigial narrow (parental combinations), 3 long narrow and 3 vestigial broad (new combinations).
4. **Observed result:** Morgan observed only parental combinations in F_2: 3/4 flies with long wings and broad abdomen, and 1/4 with vestigial wings and narrow abdomen.

Conclusion: The genes for wing length and abdomen width are located on the same chromosome (linked), so they did not assort independently.

Q.52 [Long] What is crossing over? How does it modify the progeny?

During the formation of gametes (meiosis-I), homologous chromosomes go through **crossing over**. The chromatids of two homologous chromosomes line up next to one another and exchange sections of DNA; the DNA strands actually break and re-join. After crossing over, the homologous chromosomes separate. The chromatids still have the same genes in the same order, but the alleles have been rearranged. When these recombinant chromatids are distributed to different gametes, a wide variety of gametes is produced. So crossing over leads to the recombination of linked genes and is responsible for genetic variability.

18.7 SEX DETERMINATION

Q.53 [Short] Define karyotype, autosomes and sex chromosomes.

The sex of an animal is determined during its embryonic development, in most animals on the basis of the **karyotype** (the numbers, sizes and shapes of the complete set of chromosomes). The chromosomes which are similar in female and male are called **autosomes**. One pair is different in female and male and determines the gender; these are the **sex chromosomes**.

Q.54 [Long] Explain the XX-XY system of sex determination.

In *Drosophila*, mammals and most other vertebrates, the sex chromosome pair consists of a rod-shaped X chromosome and a hook-shaped Y chromosome. Females are XX and males are XY. *Drosophila* has 3 pairs of autosomes and 1 pair of sex chromosomes; humans have 22 pairs of autosomes and 1 pair of sex chromosomes.

Females are **homogametic**: all their eggs carry one X chromosome. Males are **heterogametic**: they produce two types of sperm, half with an X chromosome and half with a Y chromosome. If an X-carrying sperm fertilizes the egg, the zygote is XX and the baby is a female. If a Y-carrying sperm fertilizes the egg, the zygote is XY and the baby is a male.

Gametes	X (sperm)	Y (sperm)
X (egg)	XX — female	XY — male

Q.55 [Short] Describe the XX–XO system of sex determination.

This system is found in some insects. For example, the grasshopper has 22 autosomes (11 pairs). The **male** has 23 chromosomes in total because its sex chromosome pair is XO, i.e., only an X chromosome is present and there is no Y chromosome. The male forms two types of sperm: half with X and half without any sex chromosome (**nullo gamete**). The **female** has 24 chromosomes; her sex chromosome pair is XX and all eggs carry one X. If an X-carrying sperm fertilizes the egg, an XX female is produced. If a nullo sperm fertilizes the egg, an XO male is produced.

Q.56 [Short] Describe the ZW–ZZ system of sex determination.

This system is common in birds. The sex chromosomes are **ZW in females** and **ZZ in males**. The female bird produces two kinds of eggs, half with Z and half with W, while all sperms of the male have Z. When a Z-carrying egg is fertilized, the offspring is male (ZZ). When a W-carrying egg is fertilized, the offspring is female (ZW).

The textbook sentence reads "When a Y carrying egg is fertilized..."; it should be "W carrying egg".

Q.57 [Short] What is the SRY gene?

In human males, there is a specific gene on the Y chromosome called SRY (Sex-determining Region Y). It produces a protein which starts the development of testes and prevents the development of female reproductive structures.

18.8 SEX LINKAGE

Q.58 [Short] What are sex-linked traits and pseudo-autosomal traits?

The traits which are controlled by alleles present on sex chromosomes are called **sex-linked traits**, and the phenomenon is called **sex linkage**. A sex-linked trait is actually X-linked, because the Y chromosome usually carries almost no functional genes. That is why the ratio of sex-linked traits is different in males and females.

The alleles of very few traits are found on both X and Y. Such traits are equally distributed among males and females and are called **X-Y-linked** or **pseudo-autosomal** traits.

Q.59 [Long] Explain the discovery of sex linkage (inheritance of eye colour in *Drosophila*).

1. In 1910, T. H. Morgan observed a mutant male *Drosophila* with **white eyes** instead of the normal red. He crossed it with a normal red-eyed female. All F_1 flies had red eyes, so red is dominant over white.
2. He crossed the F_1 red-eyed flies with each other. Out of 4252 F_2 flies, 782 (18%) were white-eyed, and **all white-eyed flies were males**.
3. To test whether a white-eyed female could exist, he test-crossed an F_1 red-eyed female with the white-eyed male. He obtained red-eyed and white-eyed males and females in a 1:1:1:1 ratio. So a female can have white eyes.
4. He mated a white-eyed female with a red-eyed male. All female offspring had red eyes and all male offspring had white eyes.

Conclusion: The gene for white eye is present on the X chromosome, and the Y chromosome does not have the allele of this gene. Morgan called such genes sex-linked genes. (W = red, w = white)

$F_1 \times F_1$	X^W (sperm)	Y (sperm)
X^W (egg)	X^WX^W red female	X^WY red male
X^w (egg)	X^WX^w red female	X^wY white male

Q.60 [Long] Describe the pattern of inheritance of X-linked recessive traits.

X-linked recessive traits are less common in females than in males. Such a trait appears in a female only when she has two recessive alleles on her two X chromosomes. In a male, it appears when only one allele is present on his X chromosome, because the Y chromosome does not carry its allele. Examples: muscular dystrophy, red-green colour blindness and haemophilia.

- If a mother has the trait, all of her sons will also have it.
- If the mother is a carrier (heterozygous), 50% of her sons can inherit it.
- If the trait is present in the father, all of his daughters will be carriers and no son will inherit it (sons do not inherit their father's X chromosome).

Q.61 [Short] Describe X-linked dominant traits.

X-linked dominant traits are in equal ratio in females and males. Examples: vitamin D-resistant rickets, Rett syndrome and Aicardi syndrome. If such a trait is present in the mother, 50% of her offspring can inherit it. If it is present in the father, all of his daughters will inherit it and no son will inherit it.

Q.62 [Short] What are Y-linked traits and holandric genes?

The Y chromosome contains relatively few genes compared with the X chromosome, so Y-linked traits are relatively rare. Y-linked genes do not have alleles on the X chromosome. Such genes are called **holandric genes** (or hemizygous genes). They are located only on the non-homologous region of the Y chromosome. Y-linked traits appear only in males and are inherited from father to son only. Examples in humans: hypertrichosis (growth of hair on the rim of the pinna), porcupine man (straight hair on the body) and webbing of toes.

Q.63 [Long] Write a note on haemophilia.

Haemophilia is a rare X-linked recessive trait. In this disease, the blood fails to clot properly after an injury. It is due to the reduction, malfunctioning or absence of blood clotting factors. It is of three types:

- **Haemophilia A** — abnormal clotting factor VIII; X-linked recessive.
- **Haemophilia B** — abnormal clotting factor IX; X-linked recessive.
- **Haemophilia C** — abnormal clotting factor XI; autosomal recessive, so present in both genders equally.

Inheritance of haemophilia A and B: Being X-linked recessive, they are more common in males than females. A male with one recessive allele (X^hY) is haemophiliac. A female is haemophiliac only when she has two recessive alleles (X^hX^h); with one allele (X^HX^h) she is a normal carrier. For example, a carrier mother and a normal father:

Gametes	X^H (sperm)	Y (sperm)
X^H (egg)	X^HX^H normal daughter	X^HY normal son
X^h (egg)	X^HX^h carrier daughter	X^hY haemophiliac son

So 50% of the sons of a carrier mother can be haemophiliac. A haemophiliac father passes the allele to all his daughters (carriers) but to none of his sons.

Q.64 [Short] If two normal parents have a haemophiliac child, what could be the genotypes of the parents?

Father is normal: X^HY. The child must have received X^h from the mother, so the mother is a carrier: X^HX^h. The haemophiliac child is a son (X^hY).

Q.65 [Short] Describe the mechanism of normal blood clotting.

1. When an injury occurs, platelets adhere to the site and form a platelet plug.
2. Activated platelets and the damaged tissue release clotting factors.
3. The clotting factors perform a clotting cascade, in which the soluble protein fibrinogen is converted into insoluble fibrin.
4. Fibrin forms a mesh that entraps parts of the blood and makes a solid clot.
5. The clot is stabilized by the cross-linking of fibrin strands.

Q.66 [Long] Write a note on colour blindness.

In people with normal (trichromatic) vision, there are three kinds of cone cells in the retina: red-sensitive, green-sensitive and blue-sensitive cones. Each contains its specific pigment protein called **opsin**. The genes for red and green opsins are on the X chromosome, while the gene for blue opsin is on autosome 7. The recessive alleles for opsins cause colour blindness.

- **Dichromacy:** perception of two primary colours and blindness of the third, whose opsin is missing. It may be red-blindness (protanopia), green-blindness (deuteranopia) or blue-blindness (tritanopia).
- **Monochromacy** (total colour blindness): perception of a single colour and blindness of the two colours whose opsins are missing. In blue-cone monochromacy, red and green opsins are absent; it is also called red–green colour blindness.

Inheritance: Red–green colour blindness is an X-linked recessive trait, so it is more common in males. A male needs only one recessive allele on his X chromosome to be colour-blind, while a female needs two. A carrier mother can pass it to 50% of her sons; a colour-blind father makes all his daughters carriers.

There is no cure. Colour-blind glasses, special software and colour discrimination training can help.

Q.67 [Short] Write a note on muscular dystrophy.

There are more than 30 muscular dystrophies; **Duchenne muscular dystrophy** is the most common. It is X-linked recessive, so more common in males. About two-thirds of cases are inherited from the mother and one-third are due to a new mutation. A male with the recessive allele cannot make the muscle protein **dystrophin**. The muscle fibres die, and there is weakness and breakdown of skeletal muscles, especially of the hips, pelvic area, thighs and calves; later of the shoulders, neck, arms, and respiratory and cardiac muscles. It affects a child around the age of four. By age 12, most patients need a wheelchair. Typical lifespans range from 15 to 45. There is no cure; physical therapy, corticosteroids, pain-killers and surgery help to manage it.

Q.68 [Short] What are sex-limited traits?

The genes of sex-limited traits are present in both sexes but are expressed in only one sex, due to anatomical or hormonal differences. Examples: genes of horn development are expressed only in male sheep; genes of milk production are expressed only in female mammals; in humans, the genes for growth of beard and for prostate cancer are expressed only in males, and the genes for ovarian cancer only in females.

Q.69 [Short] What are sex-influenced traits?

In sex-influenced traits, the same gene shows different expression in each sex. Sex-influenced genes have alleles which are dominant in one sex and recessive in the other, due to the hormonal difference between males and females. Examples in humans: amount of body hair and baldness. The allele of baldness is dominant in men but recessive in women. So a man can be bald when he is homozygous or heterozygous for the baldness allele, while a woman can be bald only when she inherits two baldness alleles.

The textbook writes that a man is bald with BB or Bb while a woman is bald only with "two recessive alleles (bb)". The two statements use the symbols differently; the idea is that a woman needs two baldness alleles.

Q.70 [Short] What is embryonic screening? Describe amniocentesis.

Embryonic screening means the methods used to check genetic disorders and abnormal numbers of chromosomes in embryos. It includes ultrasound scan, chorionic villus sampling and amniocentesis.

Amniocentesis is generally done between the 15th and 20th week of pregnancy. A long needle is inserted into the amniotic sac while an ultrasound probe locates the foetus. The needle sucks some amniotic fluid, which contains cells of the foetus. The foetal cells are kept in a culture medium and checked for an abnormal number of chromosomes.

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